Programming Exercise

Questions

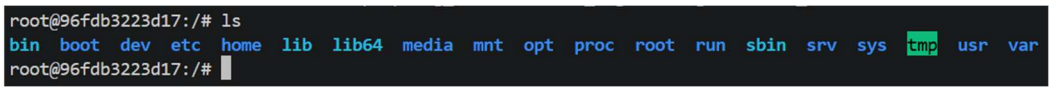
Provide a 50-to-100-word analysis for each of the questions.

1. In the program, there are two libraries imported to run docker commands from Python: “os” and “subprocess”. Explain why these libraries are needed and beneficial for docker functionality.

   The os module allows the program to check if a file is in a given directory. For this program, it is used to verify the Dockerfile is present. The subprocess module allows the program to run shell commands in Python. It is used to execute docker commands in this program, execute_command(“docker image ls”) for example. These libraries are beneficial because they allow running docker workflows within scripts instead of having to use the terminal.

2. What does the command “docker run –it <image> do”? Explain the functionality of each part of the command.

   The command docker run -it <image> creates a container from the given image and runs it. Docker creates the container, -it allows users to interact with docker through a pseudo-terminal, <image> is where the image name goes to run the proper image. Executing this command in this manner gives the user ther ability to manipulate the container in the same manner as they would be able to in a normal terminal session.

3. Show a print screen of the execution of the ls command in the last step of the code. Explain why executing this command in a terminal may be more beneficial than in the Jupyter notebook cell.

   Jupyter notebook cells are made to run lines of code individually. Running the ls command in a terminal allows the user to see the container’s file system in real time within an interactive environment. This is less realistic in notebooks due to their segregated nature. Using the terminal provides immediate feedback and a more natural way to explore and interact with the container.